Stack Simulator Pro – Program

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Stack Simulator Pro</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background: #f2f5f9;
      text-align: center;
      margin: 0;
      padding: 30px;
    }

    h1 {
      color: #1f2937;
    }

    .container {
      max-width: 600px;
      margin: auto;
      background: white;
      padding: 25px;
      border-radius: 12px;
      box-shadow: 0 4px 12px rgba(0,0,0,0.15);
    }

    input {
      padding: 12px;
      width: 200px;
      border: 1px solid #aaa;
      border-radius: 6px;
      font-size: 16px;
    }

    button {
      padding: 12px 18px;
      margin: 6px;
      border: none;
      border-radius: 6px;
      background: #2563eb;
      color: white;
      cursor: pointer;
      font-size: 15px;
    }

    button:hover {
      background: #1d4ed8;
    }

    #stack {
```

```
            margin: 25px auto;
            width: 200px;
            min-height: 250px;
            border: 3px solid #333;
            border-top: none;
            display: flex;
            flex-direction: column-reverse;
            justify-content: flex-start;
        }

        .element {
            background: #60a5fa;
            color: white;
            margin: 3px;
            padding: 12px;
            font-weight: bold;
            border-radius: 5px;
        }

        #message {
            font-weight: bold;
            margin-top: 15px;
        }
    </style>
</head>

<body>

    <div class="container">

        <h1>Stack Simulator Pro</h1>

        <p><b>Stack follows LIFO:</b> Last In, First Out</p>

        <input type="text" id="value" placeholder="Enter value">

        <br>

        <button onclick="push()">Push</button>
        <button onclick="pop()">Pop</button>
        <button onclick="peek()">Peek</button>
        <button onclick="display()">Display</button>
        <button onclick="clearStack()">Clear</button>

        <h3>Stack</h3>

        <div id="stack"></div>

        <p id="message"></p>

    </div>

    <script>
```

```
// Stack array
let stack = [];

// Push operation
function push() {

   let value = document.getElementById("value").value;

   if (value === "") {
      showMessage("Please enter a value.");
      return;
   }

   stack.push(value);

   document.getElementById("value").value = "";

   showMessage("Pushed: " + value);

   display();
}

// Pop operation
function pop() {

   if (stack.length === 0) {
      showMessage("Stack Underflow! Stack is empty.");
      return;
   }

   let value = stack.pop();

   showMessage("Popped: " + value);

   display();
}

// Peek operation
function peek() {

   if (stack.length === 0) {
      showMessage("Stack is empty.");
      return;
   }

   let top = stack[stack.length - 1];

   showMessage("Top element: " + top);
}

// Display operation
function display() {
```

```
        let stackDiv = document.getElementById("stack");

        stackDiv.innerHTML = "";

        for (let i = 0; i < stack.length; i++) {

            let element = document.createElement("div");

            element.className = "element";

            element.innerText = stack[i];

            stackDiv.appendChild(element);
        }
    }

    // Clear stack
    function clearStack() {

        stack = [];

        display();

        showMessage("Stack cleared.");
    }

    // Display messages
    function showMessage(message) {

        document.getElementById("message").innerText = message;
    }

  </script>

</body>
</html>
```